



AUTONOMOUS FRAMEWORK FOR REAL-TIME DATA QUALITY, PRIVACY, AND SCHEMA DRIFT MANAGEMENT IN STREAMING ENVIRONMENTS

Presented By: 25-26J-090

GROUP MEMBERS



AMODHYA



LAVINDI

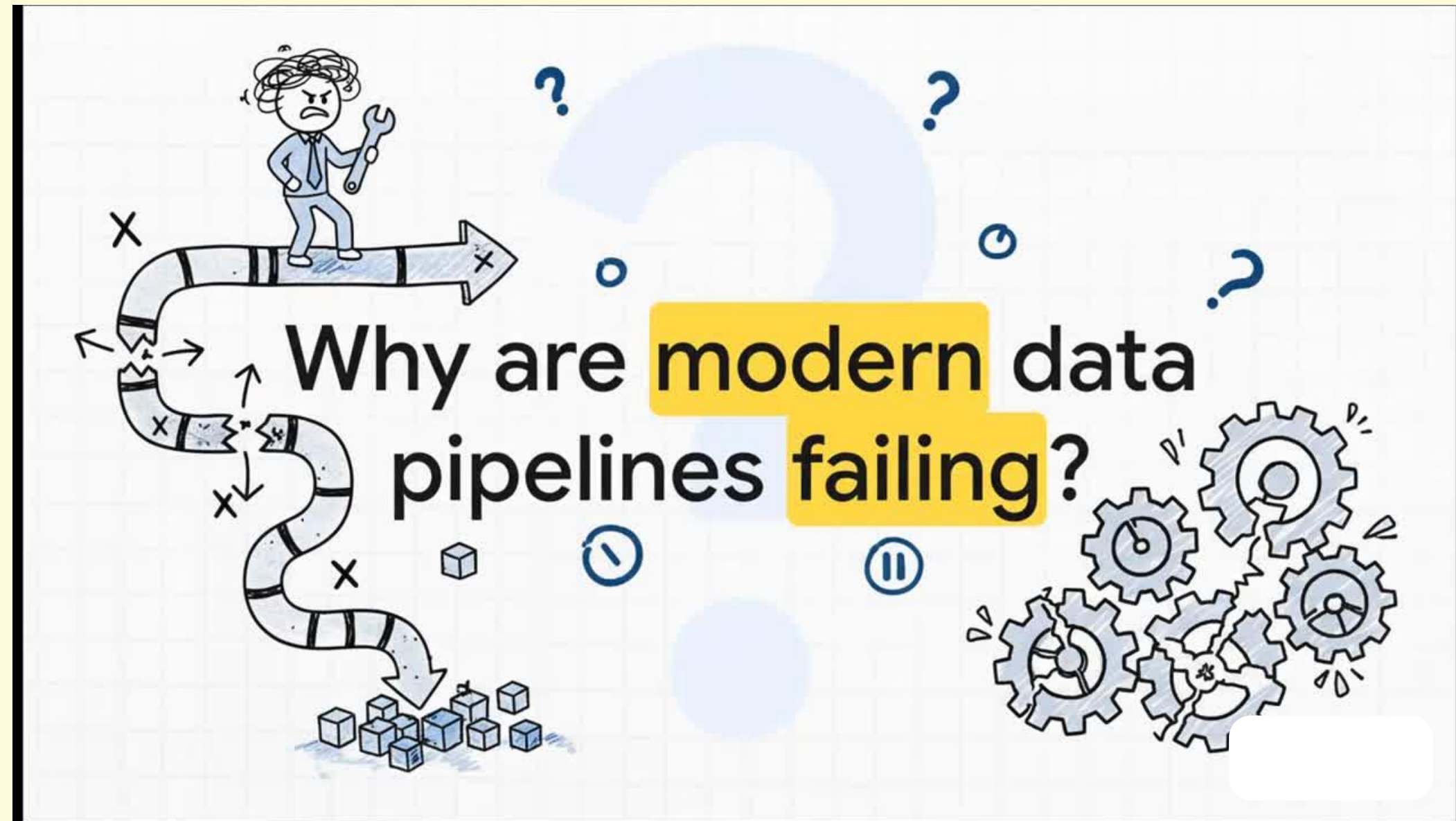


AMANDA



NANDUNI

PROBLEM DEFINITION



SILVA A.S.

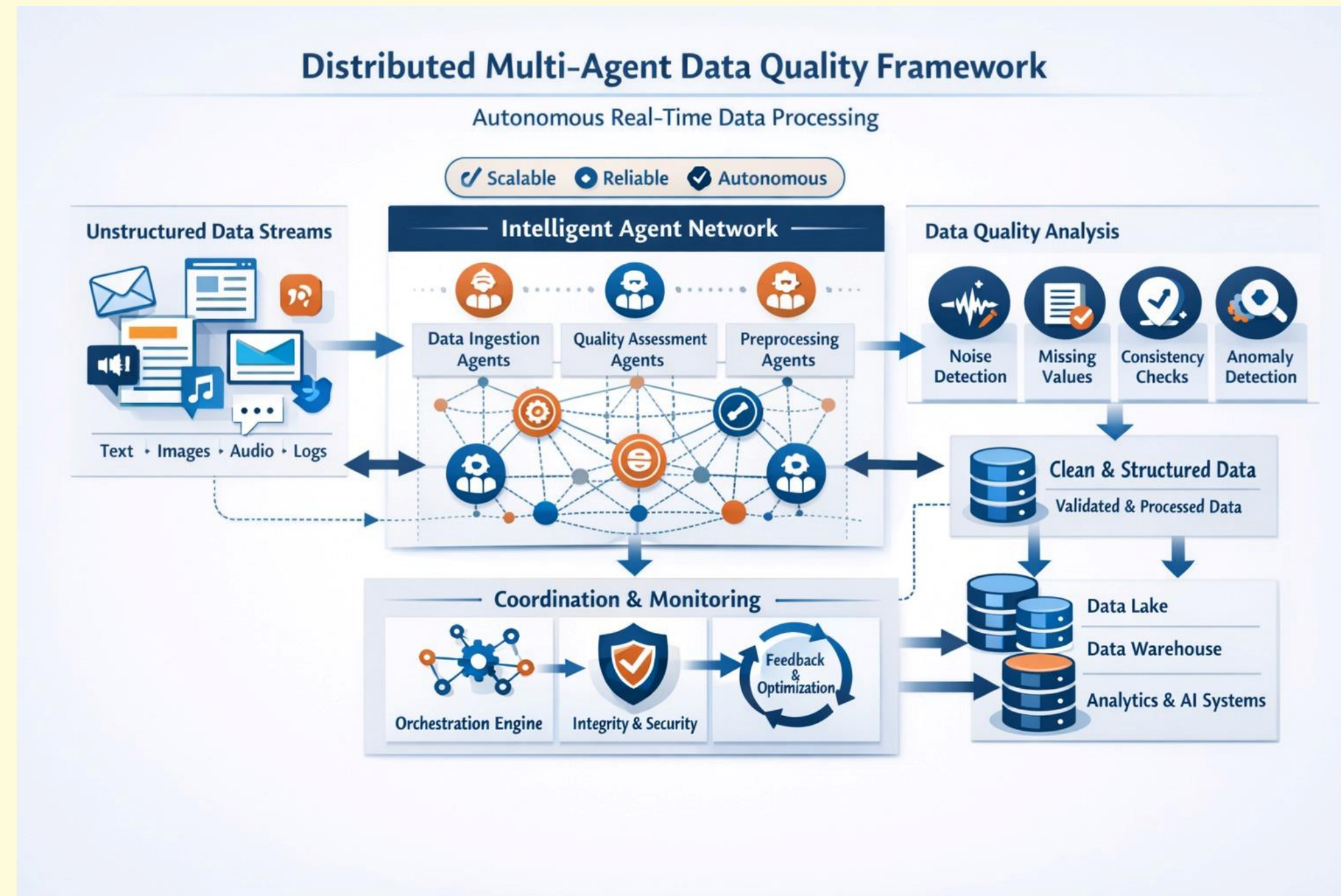
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**AUTONOMOUS MULTI-AGENT FRAMEWORK FOR REAL-TIME QUALITY
ASSESSMENT PREPROCESSING OF UNSTRUCTURED DATA STREAMS**

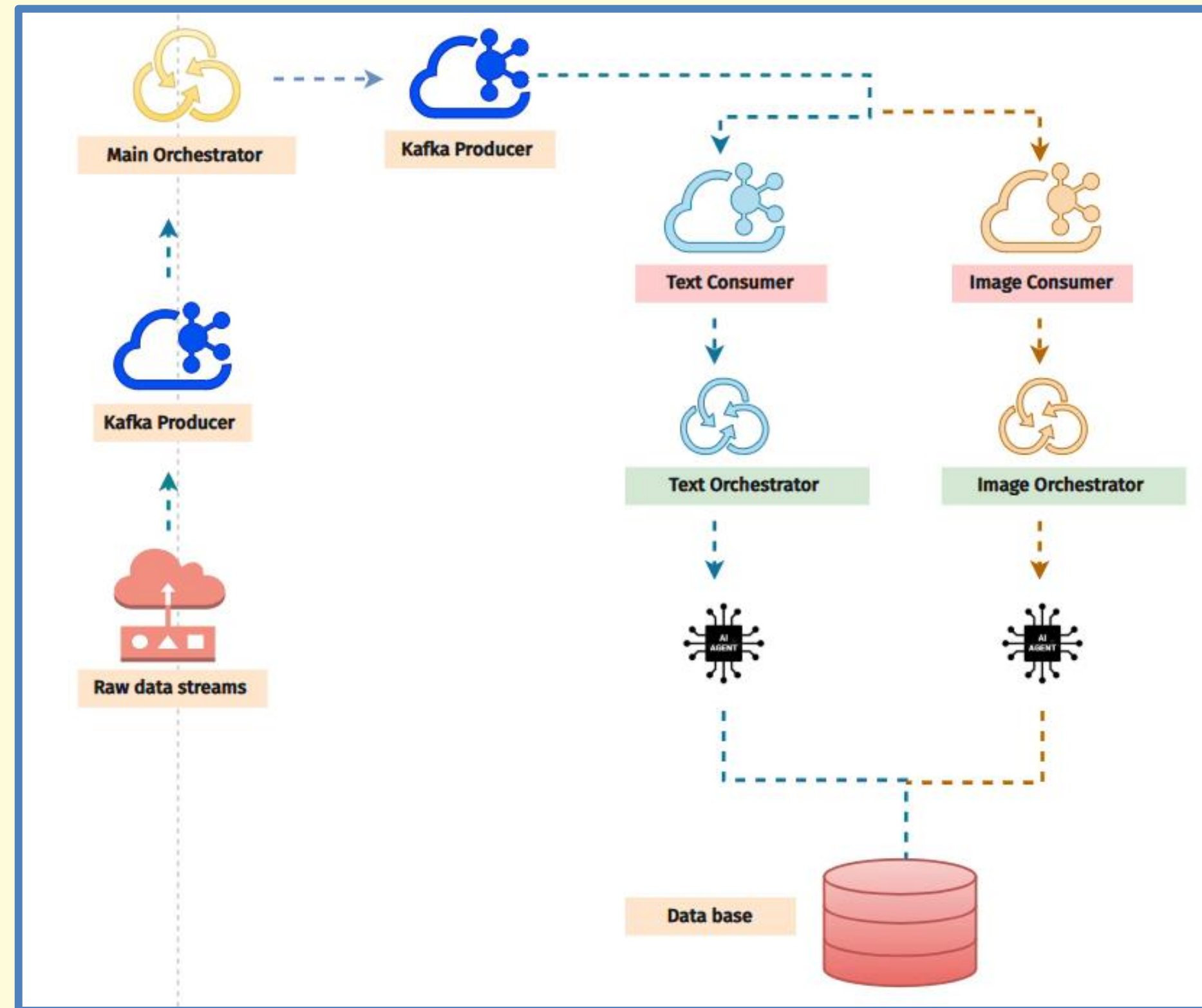
AUTONOMOUS MULTI-AGENT FRAMEWORK FOR REAL-TIME QUALITY ASSESSMENT PREPROCESSING OF UNSTRUCTURED DATA STREAMS

Problem



AUTONOMOUS MULTI-AGENT FRAMEWORK FOR REAL-TIME QUALITY ASSESSMENT PREPROCESSING OF UNSTRUCTURED DATA STREAMS

Developed
Solution



AUTONOMOUS MULTI-AGENT FRAMEWORK FOR REAL-TIME QUALITY ASSESSMENT PREPROCESSING OF UNSTRUCTURED DATA STREAMS

User Requirements vs Design Excellence

 User Requirements	 Design Excellence
 Real-time unstructured data processing	 Autonomous agent orchestration
 Zero human intervention	 Scalable distributed architecture
 High availability and resilience	 Fault-tolerant and reliable processing
 Trusted, validated outputs	 End-to-end data integrity and traceability

ARACHCHI L.T.E.

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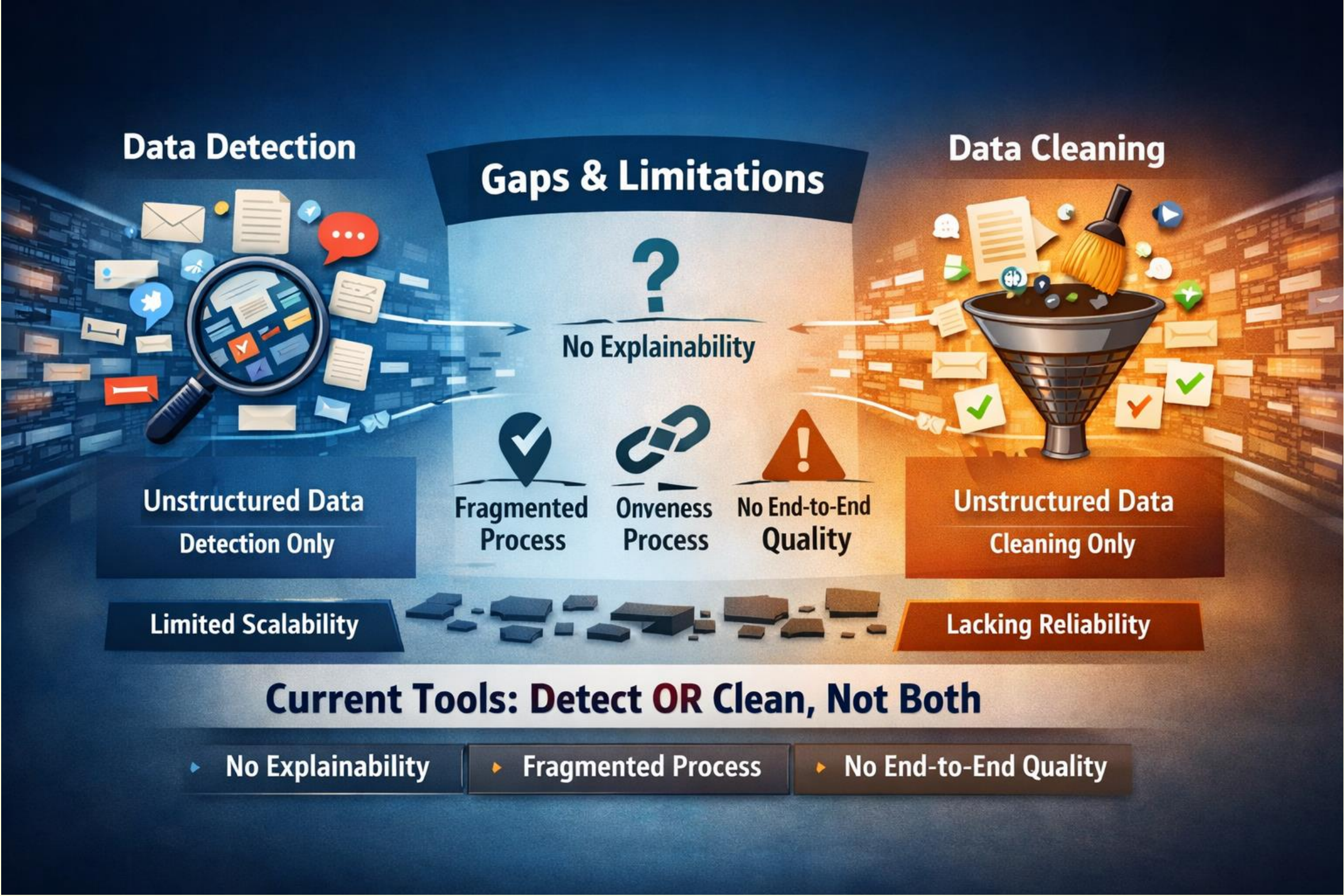


**AUTONOMUS DATA QUALITY ASSESSMENT AND EXPLAINABLE
PREPROCESSING FRAMEWORK FOR STRUCTURED DATA**

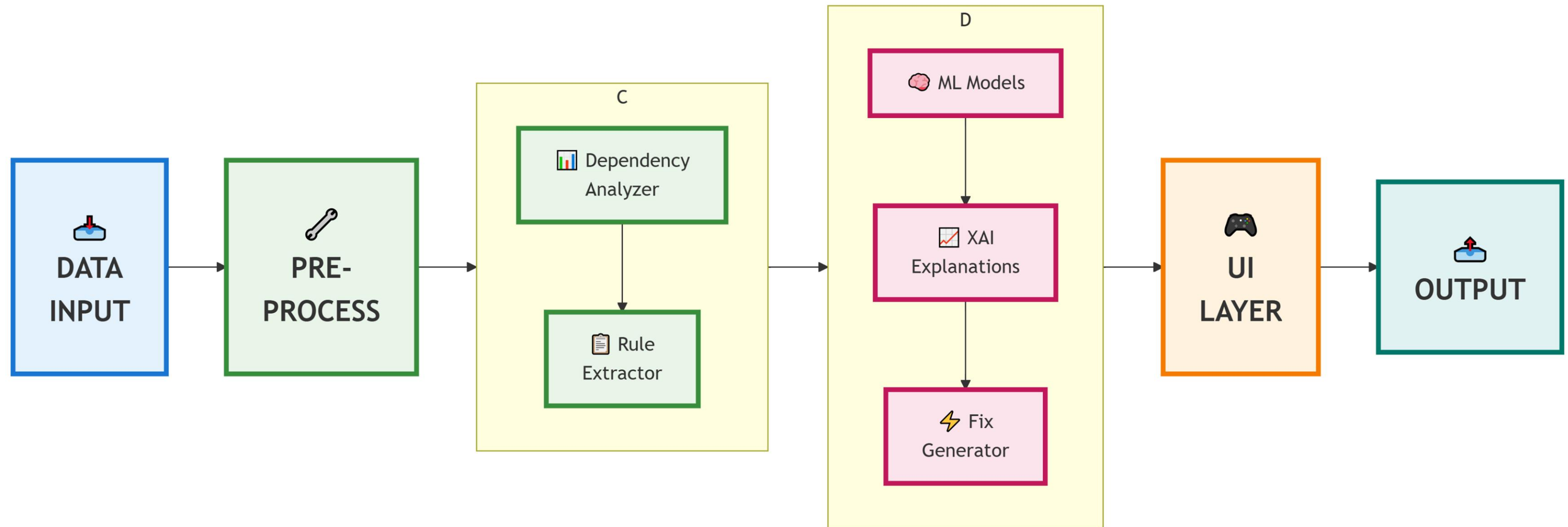
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AUTONOMUS DATA QUALITY ASSESSMENT AND EXPLAINABLE
PREPROCESSING FRAMEWORK FOR STRUCTURED DATA

Problem



SOLUTION



AUTONOMUS DATA QUALITY ASSESSMENT AND EXPLAINABLE PREPROCESSING FRAMEWORK FOR STRUCTURED DATA

Design Contributions vs User Requirements

 User Requirements	 Design Contribution
 Automatic detection of data quality issues	 Modular, extensible pipeline architecture
 Column-wise ML-based imputation	 Hybrid rules + ML approach for robustness
 Rule-based logical constraint validation	 Per-feature learning to preserve data semantics
 Transparent issue logging and fix suggestions	 Easily integrable with ML workflows

ARANGALLA H.A.A.K.

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PRIVACY PRESERVATION IN DATA PIPELINES FOR STREAMING ENVIRONMENTS

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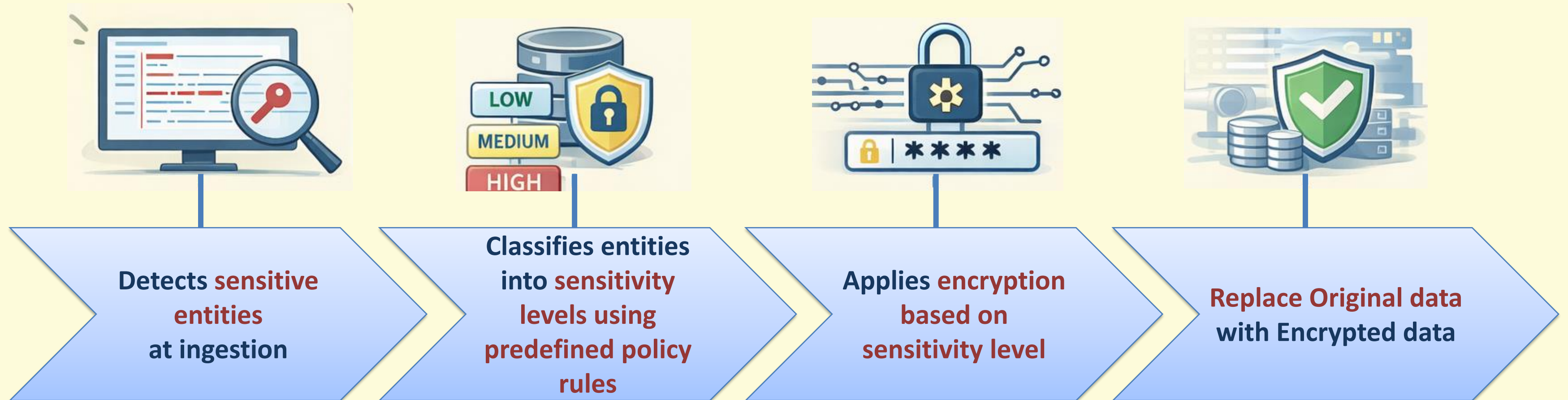
PRIVACY PRESERVATION IN DATA PIPELINES FOR STREAMING ENVIRONMENTS

Problem



PRIVACY PRESERVATION IN DATA PIPELINES FOR STREAMING ENVIRONMENTS

How the proposed solution can stop sensitive data exposure?



PASS ONLY SECURED DATA TO DOWNSTREAM SYSTEMS!!!

PRIVACY PRESERVATION IN DATA PIPELINES FOR STREAMING ENVIRONMENTS

Addressed User Requirements



- Eliminates manual identification of sensitive data



- No raw PII stored or logged



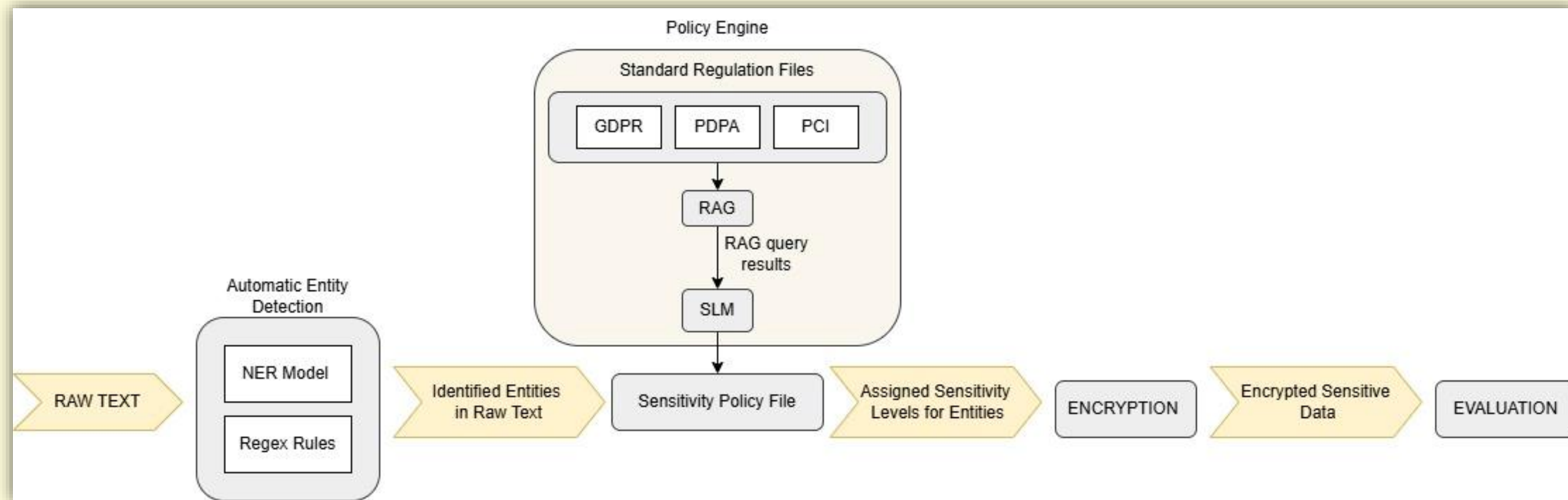
- GDPR, PDPA, PCI-DSS aligned privacy enforcement



- Organization-defined privacy rules

PRIVACY PRESERVATION IN DATA PIPELINES FOR STREAMING ENVIRONMENTS

Design Excellence / Contribution



3-phase privacy pipeline:

NER + regex-based sensitive data detection

Policy-driven sensitivity classification (RAG + SLM)

Adaptive encryption per sensitivity level

-> **Accurate** Entity Identification

-> **Maintainable & Scalable** Policy Engine

-> **Sensitivity Aware** Data Securing

WEERAKOON W.R.W.M.N.S

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AUTOMATED REAL-TIME SCHEMA DRIFT DETECTION AND ADAPTATION FRAMEWORK

AUTOMATED REAL-TIME SCHEMA DRIFT AND ADAPTATION FRAMEWORK

Problem Definition



Data schemas change unexpectedly
(drop, rename, add columns)



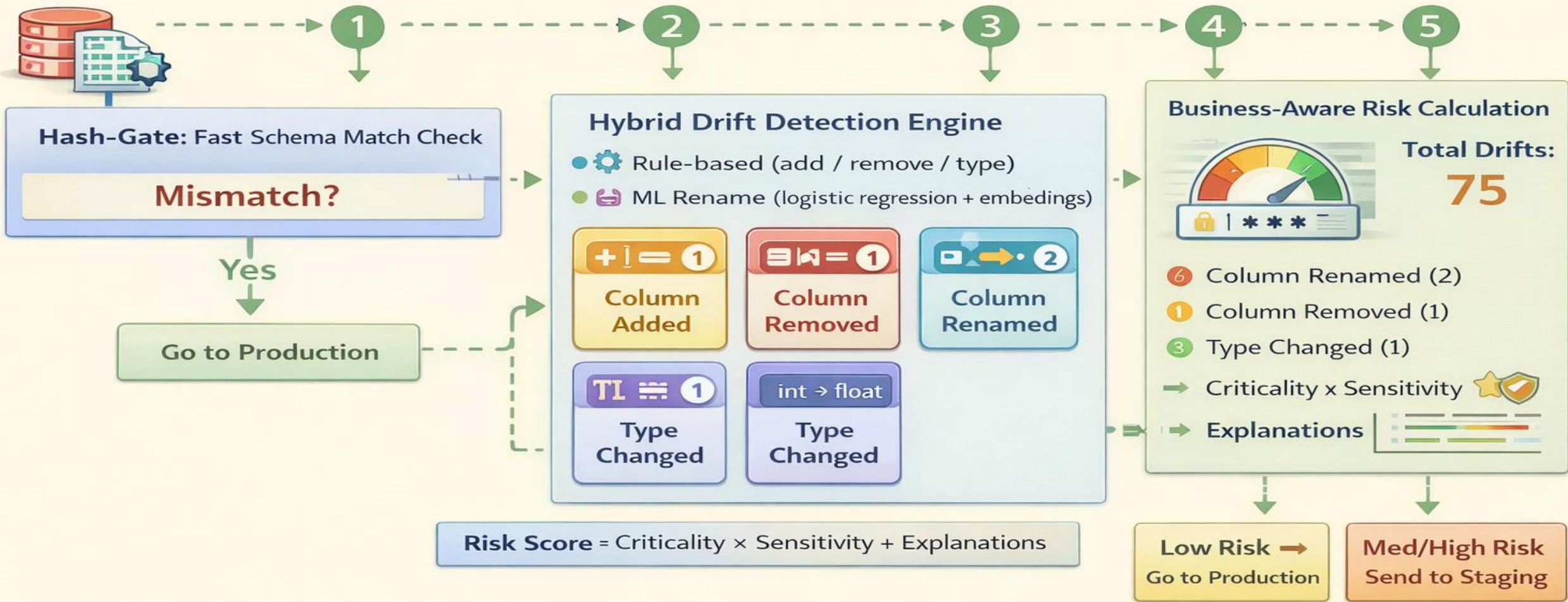
Causes pipeline failures and wrong
analytics



Manual handling is slow and error-prone

AUTOMATED REAL-TIME SCHEMA DRIFT AND ADAPTATION FRAMEWORK

How the component detects & assesses schema drifting



AUTOMATED REAL-TIME SCHEMA DRIFT AND ADAPTATION FRAMEWORK



User Requirements Addressed



Automatic schema change detection



Minimal false alerts



Safe handling of high-risk data



Clear explanations for decisions



Design Excellence



Fast schema hash comparison



Hybrid drift detection (rules + ML)



Business-aware risk scoring



Safe data routing
(production or staging)

COMMERCIALIZATION & SUSTAINABILITY

- Customer Persona



- Market Size

Rapidly growing global market for real-time analytics, streaming platforms, and data governance solutions driven by cloud and big data adoption

COMMERCIALIZATION & SUSTAINABILITY

- What is Unique



Real-time automated data correction



Agentic AI for unstructured streams



Built-in PII detection & protection



Real-time schema drift detection



One unified solution, minimal manual rules

- Cost Recovery

SaaS subscription | Enterprise licensing | Pay-per-pipeline / data volume | Support & compliance add-ons

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THANK YOU